



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 2.08.2026

In addition to Part-I (General Handout for all courses appended to the Time Table) this portion gives further specific details regarding the course.

Course No. : **CE G616**
Course Name : **BRIDGE ENGINEERING**
Instructor-in-charge : **MANOJ KUMAR**

1. Scope and Objective of the Course:

Bridges are an inseparable part of any communication network as they are the key elements in roadways and highway networks. This course intends to impart skills for analysis & design of various types of bridges. This course will mainly focus on the design of three types of bridge superstructures namely Slab culverts/bridges, T-beam bridges, and Composite bridges. The design of the bridge substructure will include the design of elastomeric bearings and the design of abutments and piers. Moreover, the course will include the design aspects of continuous bridges. At the end of this course, the students would be equipped with the complete analysis and design of slab, T-beam and composite bridges, along with the design of substructures for them.

2. Course Description:

Purpose of bridge; Classification of bridges; Characteristics of different types of bridges; Loads stresses and their combinations; Design of RCC Slab culverts, Box-culverts, T-beam bridges; composite bridges; continuous bridges; design of bearings; design of piers and abutments and their foundations.

3. Text Book (TB) and Reference Books (RB):

- TB: Krishna Raju, N. (2018), Design of Bridges, 5th Edition, Oxford & IBH Publishing Co. Pvt. Ltd., New Delhi.
- RB1: Johnson Victor, D. (2010), Essentials of Bridge Engineering, 6th Edition, Oxford & IBH Publishing Co. Pvt. Ltd., New Delhi.
- RB 2: Ponnuswamy, S., 'Bridge Engineering', 2nd Ed., 2009, McGraw-Hill, New Delhi.
- RB 1: Raina, V. K. (2014), 'Concrete Bridge Practice: Analysis, Design and Economics, Shroff Publishers, 4th Ed., 2014, Navi Mumbai.

5. Standard Specifications and code of Practice for Bridges:

1. IRC: 5-2015, Standard Specifications and code of Practice for road bridges: section I-General features of Design", Indian Road Congress.
2. IRC: 6-2017, Standard Specifications and code of Practice for road bridges: section II-Loads and Stresses", Indian Road Congress.
3. IRC: 112-2019, Code of Practice for Concrete Road Bridges', Indian Road Congress.
4. IRC: 22-2015, Standard Specifications and code of Practice for Road Bridges: section VI-Composite Construction, Indian Road Congress.
5. SP 120-2018, Explanatory Handbook to IRC:22-2015, Indian Road Congress.
6. IRC:78-2014, Standard Specifications and code of Practice for road bridges: section VII-Foundation and Substructures", Indian Road Congress
7. IRC:83-2015, Section IX Part II Standard Specifications and code of Practice for Road bridges-Elastomeric Bearings, Indian Road Congress.
8. SP-13-2004, Guidelines for Small Bridges and Culverts', Indian Road Congress.



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7. Course Plan:

Module	Lect. Nos.	Ref. to TB/code	Learning Sessions
Introduction	1-2	Ch.1	Purpose of bridges, components of bridges, selection of optimum span of bridge
Loads on Bridge Superstructure	3-5	Ch.2, IRC 6	Dead loads on bridge superstructure, types of IRC vehicular Live loads, impact factors
Design of Slab culverts / bridges	6-11	Ch. 4, IRC 112	Dispersions of IRC Class 70R tracked vehicular loads on slabs, Maximum bending moment and shear due to IRC Class 70R tracked vehicle considering dispersion of loads, critical position of IRC Class 70R wheeled and Class-A loading for moment and shear treating the loads as concentrated loads, determination of design moment and design shear force under different IRC Class-70R and Class-A loading, design of slab culvert/bridge using the limit state method.
Design of Box-culverts	12-15	Ch. 7	Loads considered in design of box-culverts, Design of Box-culverts for highways
Design of Tee-Beam Bridges	16-24	Ch. 4 & 8	Components of T-beam bridges, Design of slabs spanning in two directions for flexure (Pigeaud's Method) and shear, Design of cantilever slab panels, Live-Load distribution in longitudinal girders using Courbon's method, Guyon & Massonet method, Hendry-Jaegar method, Design of Transverse Girders
Design of Composite Bridges	25-28	Ch. 10, IRC 22	General aspects and components of composite bridges, design of laterally restrained steel girders, types and design of shear connectors, analysis and design of composite bridges for highways
Design of Bearings & Expansion joints	29-32	Ch. 16, IRC 83	Necessity of bearings, Loads acting on bearings, Types of bearings, Design of Elastomeric Bearings, Necessity and types of expansion joints
Design of Abutments & Piers	33-37	IRC 112:8	Loads and Forces considered for designing Abutment and Pier, Determination of Design Moment and forces in abutment and pier, Limit state Design of Abutment and Pier
Design of Continuous Bridges	38-42	Ch. 15	Necessity of continuous bridges, analysis and design aspects of continuous bridges

8. Evaluation Scheme:

Components	Duration	Wtg. (%)	Date and Time	Nature
Mid-Semester Test	90 Mins.	35	10/10/2026 FN2	CB
Comprehensive Examination	3 Hrs.	45	15/12/2026 AN	CB/OB
Tutorials/Seminar/Presentations/Term Projects	---	20	----	----

9. Chamber Consultation Hour: To be announced in the class.

10. Notice: Notice, if any, concerning this course will be sent to Nalanda/Institute E-mail.

11. Make up policy: Make-up will be granted only to genuine cases. For cases related to illness, proper documentary evidence is essential. Prior permission is necessary if a student is out of station on the test date.

Instructor-in-charge
CE G616